

Vladimir Grigoriants

Bioinformatician · computational biology (R, Python)

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SUMMARY

I am a bioinformatician with 3 years of experience in immunology, analysing TCR-repertoire, bulk and single-cell RNA-seq data from raw reads to publication-ready results. I performed all bioinformatic and statistical analysis for a *Frontiers in Immunology* paper on paired blood and tonsillar T-follicular cells, including an end-to-end single-cell pipeline (CellRanger, Seurat, scanpy) and integration of single-cell and bulk TCR data. I also built and validated a gene-expression classifier of induced vs. natural Tfr cells (leave-one-patient-out AUC 0.86–0.89) and carried a mouse immunisation study from FACS sorting and library preparation through to statistical analysis. I want to keep growing as a bioinformatician, including in areas beyond immunology.

SKILLS

Programming & data	Python, R, Bash, git; Linux command-line workflows, HPC cluster (SLURM), conda, reproducible analysis scripts (optparse/argparse)
Omics	NGS: bulk TCR-repertoire sequencing (MiXCR, CellRanger VDJ), bulk RNA-seq, single-cell RNA-seq and CITE-seq (CellRanger, Seurat, scanpy); familiarity with ATAC-seq, ChIP-seq and methylation data from coursework and lab exposure
Wet lab	flow cytometry and FACS sorting, antigen preparation (protein biotinylation, streptavidin conjugation), RNA extraction and TCR-immunosequencing library preparation, recombinant protein production
Single-cell analysis	QC and normalisation, highly-variable-gene selection, batch integration (Harmony), Leiden/Louvain clustering, marker-gene-based cluster annotation (Seurat `FindAllMarkers`), cross-cohort reference mapping and co-embedding of independent datasets (up to ~100k cells across 3 cohorts); trajectory/lineage inference and pseudotime (Slingshot, Monocle3), RNA velocity estimation (velocyto)
Bulk TCR-seq analysis	clonotype calling and clustering (exact V/J/CDR3 identity and relaxed CDR3 Hamming-distance clustering), clonotype-sharing/overlap analysis between populations and donors, repertoire-diversity metrics (D50, Chao1, Shannon), donor- and population-aware downsampling for fair comparison, antigen-driven clonal-expansion detection with the ALICE method (Pogorelyy et al.)
Single-cell/bulk TCR integration	joining single-cell and bulk TCR-repertoire data by shared clonotype identity to transfer population labels onto single-cell clusters, then training an expression-based classifier that scores all single cells, including ones without a captured TCR
Statistics & ML	exact and permutation tests for small samples; stratified analysis (Cochran–Mantel–Haenszel, van Elteren); multiple-testing correction (Benjamini–Hochberg FDR, Bonferroni); effect sizes and confidence intervals (Cohen's d, Cliff's δ, Hodges–Lehmann, odds ratio, bootstrap/Wilson CIs); power and sample-size analysis; ANOVA, multiple regression; logistic regression, generalized linear mixed models, elastic net, XGBoost, SVM; leave-one-patient-out and nested cross-validation, ROC/AUC; PERMANOVA/PERMDISP, Dirichlet-multinomial mixture models, PCA/PCoA/NMDS
R packages	tidyverse (dplyr, tidyr, stringr), data.table, ggplot2, patchwork, ggrepel, ComplexHeatmap, pheatmap, plotly; lme4, glmnet, pROC, limma, xgboost, e1071; Seurat, Bioconductor
Python packages	pandas, NumPy, SciPy, statsmodels, scikit-learn, matplotlib, seaborn, plotly, scanpy

EXPERIENCE

Data Analyst — Sep 2023 – present
RNIMU n.a. N.I. Pirogov, Institute of Translational Medicine — D.M. Chudakov laboratory, molecular technologies — Moscow

Role scope: Lead statistical and bioinformatic analysis of NGS data (TCR-repertoire, bulk and single-cell RNA-seq) from human-donor and mouse experiments for an immunology research group, from exploratory analysis and model development through publication-ready figures. Act as the lead analyst for two parallel projects, taking study data from raw sequencing output to statistical inference and manuscript-ready results, working closely with the PIs (D. Chudakov and I.L. Grigorova) throughout.

- **Led the statistical and bioinformatic analysis for a paper on paired blood and tonsillar T-follicular cells,** published in *Frontiers in Immunology*, covering TCR-repertoire profiling, group comparisons, stratified tests and effect sizes for a human-donor cohort.
- **Quantified clonotype sharing between blood and tonsil T-cell populations** (four blood Tfh/Tfr subsets vs. tonsillar GC Tfh and Tregs across donors), building overlap heatmaps, replica-overlap and D50-diversity comparisons, with population sizes matched by donor- and population-aware downsampling.

- **Built the single-cell pipeline end-to-end, from raw reads to annotated clusters:** fixed a library-prep read-orientation artefact with cutadapt, ran `cellranger multi` (5′ GEX + VDJ-T, auto-detected SC5P-PE chemistry) on SLURM to produce per-donor gene-expression and TCR calls, then loaded, QC'd (24,637 cells passing QC across 2 donors) and clustered the data in Seurat/R.
- **Processed and annotated single-cell RNA/CITE-seq data in Seurat and scanpy:** QC and normalisation, HVG selection, Harmony batch integration and Leiden clustering, co-embedding the project's single-cell reference with external public cohorts (up to ~100,000 cells across 3 cohorts) for joint cluster annotation (marker-gene `FindAllMarkers`, manual curation) and TCR-sharing overlay per joint cluster.
- **Integrated single-cell and bulk TCR-repertoire data by shared clonotype identity** (exact V/J/CDR3 match, plus a relaxed CDR3 Hamming-distance clustering step) to transfer bulk population labels (GC Tfh- vs. Treg-derived) onto single-cell clusters, using the TCR-based labels as ground truth for the gene-expression classifier below.
- **Built and validated a differential-expression gene signature and logistic-regression classifier for induced- vs. natural-Tfr identity from single-cell RNA-seq data**, trained against the TCR-derived population labels and scoring the full single-cell Tfr population (including cells lacking a captured TCR); evaluated with leave-one-patient-out cross-validation (AUC 0.86–0.89), benchmarked against elastic-net, XGBoost and SVM alternatives, and externally validated on an independent public cohort with a Cochran–Mantel–Haenszel pooled odds ratio.
- **Ran multi-method trajectory inference in 3D UMAP space (Slingshot, cross-checked with Monocle3)** to test competing root scenarios and support distinguishing induced- from natural-Tfr origin along the inferred lineage trajectories, as reported in the published paper's Methods.
- **Traced induced-Tfr (iTfr) identity beyond the T-follicular response,** as a follow-on research question beyond the published Frontiers in Immunology paper: used bubble plots and clonotype-sharing fold-enrichment statistics (clonotype- and cluster-level) to characterise the dual overlap between GC Tfh, single-cell Tfr and non-follicular Tregs, benchmarked against an independent healthy-donor tonsil cohort (Brenna et al.) and against non-follicular conventional T cells as a negative control, with CD38 as a marker enriching for induced Tfr in circulation.
- **Reproduced the dual-sharing enrichment pattern in an independent public mouse dataset** (Ritvo et al., PNAS 2018, NOD mouse spleen TCR-β repertoires): reprocessed the raw reads with MiXCR and quantified fold enrichment of GC Tfh clonotype sharing with Tfr + non-follicular Treg versus Tfr + non-follicular conventional T cells.
- **Applied the ALICE method** (neighbour-based antigen-specific-expansion detection, Poisson null model with BH correction) to flag antigen-experienced, clonally expanded single-cell clusters ahead of the sharing and classifier analyses.
- **Ran a mouse immunisation study end-to-end, wet lab through bioinformatics** (regulatory T-cell specificity to nuclear autoantigens): prepared the antigens, flow-sorted the target cell populations by FACS, and built the TCR-immunosequencing libraries myself, then analysed the data with small-sample statistics (animal as the independent unit: exact Mann–Whitney tests, Cliff's δ and Hodges–Lehmann estimates, mouse-level permutation tests) and ran differential-expression analysis on single-cell RNA-seq data in Seurat.

Industrial Project 2023
MiLaboratory, Skoltech MSc industrial-project track Moscow

Role scope: Automated quality control for immunosequencing reagent kits.

- Automated the QC pipeline used to validate reagent-kit batches for immunosequencing.

Research Trainee Sep 2021 – Jul 2022
Shemyakin–Ovchinnikov Institute of Bioorganic Chemistry (IBCh RAS), biocatalysis laboratory Moscow

Role scope: Supported a biocatalysis research group with wet-lab protein production.

- Produced recombinant proteins as part of the lab's ongoing research.
- Gained hands-on exposure to how experimental sequencing data is generated, which now informs how I design downstream statistical analyses.

PROJECTS

TCR clonotyping of paired blood and tonsillar T-follicular cells (Paper) Grant RSF 24-15-00545

- Co-author (Grigoriants V.A., Sokolova S.R.) of "TCR clonotyping of paired blood and tonsillar T-follicular cells reveals the mixed origin of CD38high Tfrs," *Frontiers in Immunology*; performed all bioinformatic and statistical analysis and data visualisation for the manuscript.
- Analysed human-donor TCR-repertoire and single-cell transcriptomic data with group-level nonparametric tests (Mann–Whitney, Fisher's exact, hypergeometric) under Benjamini–Hochberg FDR control, and ran a donor-stratified van Elteren analysis in response to peer review.
- Carried out power analysis, minimum-detectable-effect estimation, TOST equivalence testing, Cliff's δ, Hodges–Lehmann estimation and bootstrap confidence intervals to support a reviewer-requested revision.

> **Tracing induced-Tfr identity outside the T-follicular response (ongoing research)**
>
> - Follow-on analysis from the same blood/tonsil TCR-repertoire and single-cell dataset, not included in the published Frontiers in Immunology paper, examining where induced Tfr (iTfr) cells originate outside the germinal-centre T-follicular response.
> - Built bubble plots and fold-enrichment statistics (clonotype- and cluster-level) for dual clonotype sharing between tonsillar GC Tfh, single-cell Tfr clusters and non-follicular (blood) Tregs, with CD38 marking induced Tfr enriched in circulation.

- Benchmarked the sharing pattern against an independent published tonsil GC-Tfh cohort (Brenna et al., 7 donors) and against non-follicular conventional T cells as a negative-control population; statistical testing with Mann–Whitney U and Benjamini–Hochberg correction in Python and R.
- Reproduced the same dual-sharing enrichment (GC Tfh ∩ Tfr ∩ non-follicular Treg vs. GC Tfh ∩ Tfr ∩ non-follicular conventional T cells) in an independent public mouse TCR-repertoire dataset (Ritvo et al., *PNAS* 2018, NOD mouse spleen): downloaded the raw reads, reprocessed them with MiXCR, and computed the triple-overlap fold enrichment per mouse.

Regulatory T-cell specificity to nuclear autoantigens (MSc thesis) Best Thesis award

- Conducted an end-to-end mouse immunisation study (regulatory T-cell specificity to nuclear autoantigens) myself, from wet lab to bioinformatics, continued as part of ongoing PhD work; 40+ mice and 200+ TCR-repertoire libraries analysed in total across cohorts.
- Ran the full wet-lab pipeline: biotinylated and streptavidin-conjugated the nuclear-protein antigens, flow-sorted the activated Tfr cell populations by FACS, and prepared the RNA and TCR-immunosequencing libraries from the sorted cells.
- Built exact and permutation-based small-sample tests with mouse-level resampling, implemented in Python, and clustered TCR clonotypes by V-gene-matched CDR3 similarity (tcrGrapher) to identify public, antigen-enriched TCR clusters.
- Built a logistic-regression Treg-vs-Tconv classifier from TCR sequence alone (TRBV usage and CDR3 physicochemical features, no generation-probability feature), trained on an external MHC-matched public reference repertoire and cross-validated to establish a ceiling accuracy for per-clone identity calls, then scored every single-cell TCR clonotype in the mouse dataset with it as a cross-check against a separate k-NN-based (TCRdist3) scoring method.
- Awarded "Best Thesis among Molecular Biologists," Department of Life Sciences, Skoltech (2024).

EDUCATION

Skolkovo Institute of Science and Technology (Skoltech) — PhD, Life Sciences Sep 2024 – present

3rd-year PhD student, immunology and computational biology. Thesis topic: follicular regulatory T cells and their specificity to nuclear autoantigens.

Skoltech — MSc, Biotechnology (Life Sciences), honours Sep 2022 – Jun 2024

GPA 3.9/4, taught in English. **Applied Biostatistics for Life Sciences (D.D. Pervouchine), grade A**: study design and bias, probability and sampling distributions, estimation and confidence intervals, hypothesis testing, sample-size and power calculation, diagnostic-test evaluation, parametric and nonparametric tests, multiple comparisons, regression and ANOVA; R-based coursework. Other courses: Omics Data Analysis (A), Computational Methods in Genetics (A), Bioinformatics (A), Intro to Programming for Biologists (A). Thesis: "Identification of mouse follicular regulatory T-cell clones specific to nuclear-protein autoantigens" (A).

Bioinformatics Institute — Postgraduate Diploma, "Bioinformatics for Biologists" Sep 2023 – Jun 2024

406 h. **Statistics and Data Analysis in R (excellent)**, Introduction to Machine Learning (excellent), Discrete Mathematics (excellent), Python programming, Linux, practical NGS data-analysis workshop. Capstone project: empty-droplet detection in scRNA-seq data.

Mendeleev University of Chemical Technology — BSc, Biotechnology, honours Sep 2018 – Jul 2022

GPA 3.96/4. **Probability Theory and Mathematical Statistics (excellent)**, Mathematics, 15 credits (excellent), Modelling of Chemical-Technological Processes in Biotechnology (excellent), Computational Mathematics.

PUBLICATIONS, GRANTS & AWARDS

- **Paper:** Grigoriants V.A., Sokolova S.R. "TCR clonotyping of paired blood and tonsillar T-follicular cells reveals the mixed origin of CD38high Tfrs," *Frontiers in Immunology*. All bioinformatic and statistical analysis and data visualisation by me.
- **RSF grant 24-15-00545** (2024–2027), team member: statistical and bioinformatic analysis of human and mouse data.
- Best Thesis among Molecular Biologists, Department of Life Sciences, Skoltech (2024).
- Winner, bioinformatics and oncology hackathon, Sechenov University (2023).
- Skoltech scholarships (Academic Excellence, Champions track); Bioinformatics Institute grant (80% tuition).

LANGUAGES

Russian — native · Armenian — native · English — advanced (MSc coursework taught in English)